
EE 604P: Practicum on Advanced Electric Drives

Course Code	EE 604P
Course Name	Practicum on Advanced Electric Drives
L-T-P-C	0-0-2-1
Prerequisites	EE508 - Fundamentals of Electric Drives, EE508P - Practicum on Electric Drives or Equivalent
Intended for	M.Tech in Power Electronics and Drives (PED)
Distribution	Core for M.Tech in Power Electronics and Drives (PED)
Approval	13th Senate (Jun 2017)

Course Contents

This is a laboratory course with 3-hour sessions per week. Following is the tentative structure of the course.

- **Part 1: Predesigned experiments on the following topics**
 - **1. Induction Motor Drives** (9 hours)
 - * Field oriented control of induction motor, Direct torque/flux control of induction motor, Effect of parameter variation on the above control performance, Power
 - * Failure Ride-through of induction motor drive, Sensorless control of induction motor drive methods of speed estimation
 - **2. Synchronous Motor Drives** (9 hours)
 - * Vector control of synchronous motor drive, Controller design
 - **3. Other Motor Drives** (6 hours)
 - * Vector control of PMSM, Closed loop control of BLDC drive, Closed loop control of SRM drive
- **Part 2: Design Project:** Sizing and controller design for a closed loop drive for a given application - This will be in form of a course project (9 hours)

References

1. W. Leonhard, **Control of Electrical Drives**, Springer-Verlag, 2001.
2. Mohan N., Undeland T. M. and Robbins W. P., **Power Electronics - Converters, Applications and Design**, 3rd Edition, Wiley India, 2008.
3. Bose B. K., **Power Electronics and Variable Frequency Drives - Technology and Applications**, IEEE Press, Standard Publisher Distributors, 2001.
4. Rashid M., **Power Electronics - Circuits, Devices and Applications**, 3rd Edition, Pearson Education.
5. Krause, P. C., Wasynczuk, O., Sudhoff, S. D., **Analysis of Electric Machinery and Drive Systems**, Wiley-Interscience.